

Standard enthalpy changes of combustion and formation (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- (R1.2.3) Deduce equations and solutions to problems involving enthalpy changes of combustion and formation.
- (R1.2.4) Calculate enthalpy changes of a reaction using $\Delta H_f^\ominus$ data or $\Delta H_c^\ominus$ data:

$$\Delta H^\ominus = \Sigma(\Delta H_f^\ominus \text{ products}) - \Sigma(\Delta H_f^\ominus \text{ reactants})$$
$$\Delta H^\ominus = \Sigma(\Delta H_c^\ominus \text{ reactants}) - \Sigma(\Delta H_c^\ominus \text{ products})$$

For every chemical process and many physical changes (such as change of state) there is an associated enthalpy value. For example, there are enthalpies of neutralisation, enthalpies of combustion and enthalpies of vaporisation (for liquids transitioning to the gaseous state). Enthalpies of reaction are very important for chemists as a way to understand chemical stability and the tendency for the reaction to occur. Two such important examples are standard enthalpy changes of combustion, $\Delta H_c^\ominus$, and formation, $\Delta H_f^\ominus$. For these enthalpies, the values are determined under very specific conditions using apparatus similar to that shown in **Figure 1**.

Why do you think that it is so important to control the conditions when determining the standard enthalpies of formation and combustion?

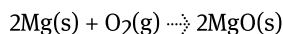


Figure 1: Regulators used to control the atmospheric conditions of an experiment.

Credit: genkur, Getty Images

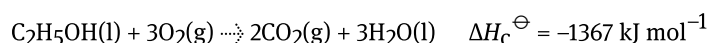
Standard enthalpy changes of combustion, $\Delta H_c^\ominus$

Combustion can be defined as an exothermic chemical process where a substance is combined with oxygen gas. Hydrocarbon combustion is a common example for combustion reactions, but metals can combust, too! For example, magnesium metal combusts to form magnesium oxide:



You will learn much more about combustion in [section R1.3.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/combustion-reactions-id-43476/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/combustion-reactions-id-43476/>). In this section we will focus on using standard enthalpies of combustion in Hess's law questions.

The molar enthalpy of combustion, or the standard enthalpy of combustion ($\Delta H_c^\ominus$), for a substance is defined as the enthalpy change when one mole of a substance is burned completely in oxygen under standard state conditions. The thermochemical equation for the combustion of ethanol is shown below:



Understanding, measuring, and comparing standard enthalpies of combustion are very useful in evaluating the efficiency of fuels and determining the calories in food.

Nature of Science

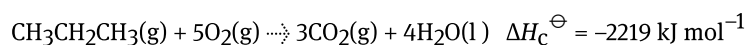
Aspect: Science as a shared endeavour

Climate change with respect to combustion of hydrocarbons

Science drives a portion of governmental policy, and when communicated effectively it has the power to shift the attitude and behaviour of societies. We have seen this recently with societal shifts from the COVID pandemic and climate change. Since both the pandemic and climate change are worldwide issues, it highlights the importance of collaboration. How can we support and improve collaboration worldwide to support progress toward shared goals?

Standard enthalpy of combustion values are found in [section 14](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/enthalpies-of-combustion-id-45116/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/enthalpies-of-combustion-id-45116/>) of DP chemistry data booklet and refer to values determined at a temperature of 298.15 K and a pressure of 100 kPa.

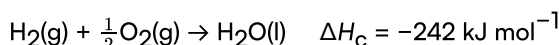
Another example of a thermochemical equation is shown. This equation is for the complete combustion of one mole of propane, $\text{CH}_3\text{CH}_2\text{CH}_3$.



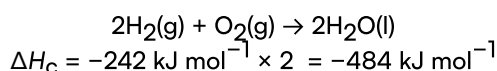
Study skills

It is important to note that the values that you find in section 14 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/enthalpies-of-combustion-id-45116/>) of the DP chemistry data booklet are for one mole of the substance that is combusted. Sometimes this requires fractions when balancing the equation. In order to arrive at integer coefficients, it is important to remember that you need to multiply the enthalpy value by the same number that you used to multiply the coefficients.

Consider this example for the combustion of hydrogen gas:

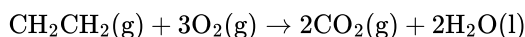


Note that $\frac{1}{2}\text{O}_2$ was required to balance this equation so that there was one mol of H_2O . In order to arrive at only integers for the coefficients, the entire equation needs to be multiplied by two — including the enthalpy value:



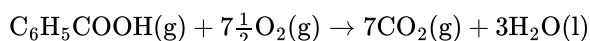
Worked example 1

Write the balanced equation for the combustion for ethene (CH_2CH_2).



Worked example 2

Write the balanced equation for the combustion for benzoic acid ($\text{C}_6\text{H}_5\text{COOH}$).



Standard enthalpy changes of formation, $\Delta H_{\text{f}}^{\ominus}$

The molar enthalpy of formation or the standard enthalpy of formation ($\Delta H_{\text{f}}^{\ominus}$) for a substance is defined as the energy change that occurs when one mole of a substance is formed from its constituent elements in their standard state.

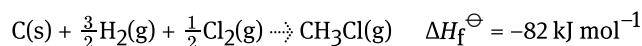
The constituent elements of a compound must be included in the state at which they exist under standard conditions. Most elements are solids under standard conditions with the exception of eleven elements which are gases and two which are liquids (Hg and Br_2). Importantly, some of these gases and liquids must always be written in their diatomic form (H_2 , Cl_2 , Br_2 , etc).

Making connections

Allotropes of carbon (diamond, graphite, fullerenes, and graphene) are all technically carbon in its standard state (see section S2.2.7 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/standard-enthalpy-changes-of-combustion-and-formation-hl-id-45309/print/>)).

[753301/book/covalent-network-structures-id-43593/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/covalent-network-structures-id-43593/)). Would you expect allotropes of an element, such as diamond and graphite, to have different $\Delta H_f^\ominus$ values?

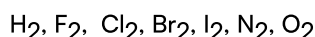
Standard enthalpy of formation values are found in [section 13](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/>) of the DP chemistry data booklet. Like standard enthalpies of combustion, this information can be conveyed in an equation together with the change in enthalpy.



Study skills

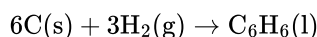
As was the case for standard enthalpies of combustions, the values in [section 13](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/>) of the DP chemistry data booklet are for the formation of one mol of the product. Sometimes this requires fractions to be used in the balancing.

It is also important to note that all reactants must be written as elements in their standard state. Typically this will be a single atom, except for diatomic elements:



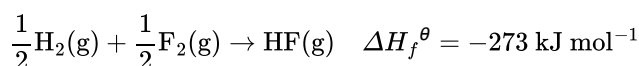
Worked example 3

Write the balanced equation for the formation of benzene (C_6H_6) from its elements.



Worked example 4

Write the balanced equation for the formation of HF, stating $\Delta H_f^\ominus$ as found in the DP chemistry data booklet in [section 13](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/>).



Study skills

Be really careful when choosing values from [section 13](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/>) of the DP chemistry data booklet as the magnitude of the enthalpy of formation is dependent on the state of the molecule.

For example, the standard enthalpy of formation for liquid water is -286 kJ mol^{-1} while the standard enthalpy of formation for gaseous water (steam) is -242 kJ mol^{-1} .

How can you explain the difference in magnitude for the enthalpies of formation for liquid versus gaseous water?

Hess's law and standard enthalpies of combustion

Calculating standard enthalpy changes of combustion, $\Delta H_c^\ominus$, and formation, $\Delta H_f^\ominus$ is an application of Hess's law. Therefore, standard enthalpies of combustion or formation can be used to calculate the enthalpy of reaction using an energy cycle like what we saw in [section R1.2.2 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/hesss-law-id-46252/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/hesss-law-id-46252/).

For combustion, we can look at this using the following equation where the enthalpy of reaction is the difference between the sum of reactant enthalpies of combustion and the sum of the product enthalpies of combustion. Importantly, each enthalpy must be multiplied by the stoichiometric balancing term (coefficient) in the balanced equation before the enthalpies are summed.

$$\Delta H^\ominus = \Sigma(\Delta H_c^\ominus \text{ reactants}) - \Sigma(\Delta H_c^\ominus \text{ products})$$

Below is an example for determining the enthalpy for the formation of glucose using the standard enthalpies of combustion (from [section 14 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/enthalpies-of-combustion-id-45116/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/enthalpies-of-combustion-id-45116/) of the DP chemistry data booklet) of glucose and its constituent elements. This example is presented in **Table 1** to assist with organising the data.

Table 1. Determining the enthalpy for the formation of glucose.

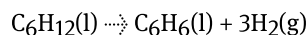
	6C(s)	+	6H ₂ (g)	+	3O ₂ (g)	→	C ₆ H ₁₂ O ₆ (s)
	$\Delta H^{\ominus} = ?$						
$\Delta H_{\text{c}}^{\ominus}$ (kJ mol ⁻¹)	-394		-286		0		-2803
Coefficient	6		6		3		1
Total enthalpy (kJ mol ⁻¹)	= (-394) × 6 = -2364		= (-286) × 6 = -1716		= (0) × 3 = 0		= (-2803) × 1 = -2803
Sums (kJ mol ⁻¹)	$\Sigma(\Delta H_{\text{c}}^{\ominus} \text{ reactants})$ = (-2364) + (-1716) + (0) = -4080						$\Sigma(\Delta H_{\text{c}}^{\ominus} \text{ products})$ = (-2803) = -2803

$$\begin{aligned}\Delta H^\ominus &= \Sigma(\Delta H_c^\ominus \text{ reactants}) - \Sigma(\Delta H_c^\ominus \text{ products}) \\ &= (-4080) - (-2803) \\ &= -1277 \text{ kJ mol}^{-1}\end{aligned}$$

Worked example 5

Using the standard enthalpies of combustion from [section 14](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/enthalpies-of-combustion-id-45116/>) of the DP chemistry data booklet, calculate the enthalpy of reaction for the desaturation of cyclohexane to benzene:



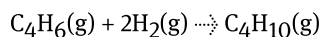
	$\text{C}_6\text{H}_{12}(\text{l})$	$\rightarrow$	$\text{C}_6\text{H}_6(\text{l}) + 3\text{H}_2(\text{g})$
$\Delta H_c^\ominus$ (kJ mol^{-1})	-3920		-3
Coefficient	1		
Total enthalpy (kJ mol^{-1})	-3920		-3
Sums (kJ mol^{-1})	$\Sigma(\Delta H_c^\ominus \text{ reactants})$ $= -3920$		$\Sigma(\Delta H_c^\ominus \text{ products})$ $= (-3920) + (-4126)$ $= -8046$

$$\begin{aligned}\Delta H^\ominus &= \Sigma(\Delta H_c^\ominus \text{ reactants}) - \Sigma(\Delta H_c^\ominus \text{ products}) \\ &= (-3920) - (-8046) \\ &= +4126 \text{ kJ mol}^{-1}\end{aligned}$$

Worked example 6

Using the standard enthalpies of combustion from [section 14](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/enthalpies-of-combustion-id-45116/>) of the DP chemistry data booklet, calculate the enthalpy of reaction for the hydrogenation of buta-1,3-diene:



	$\text{C}_4\text{H}_6(\text{g})$	$+$	$2\text{H}_2(\text{g})$
$\Delta H_c^\ominus$ (kJ mol^{-1})	-2541		-2
Coefficient	1		
Total enthalpy (kJ mol^{-1})	-2541		-2

	$\text{C}_4\text{H}_6(\text{g})$	+	$2\text{H}_2(\text{g})$
Sums (kJ mol^{-1})	$\Sigma(\Delta H_c^\ominus \text{ reactants})$ $= (-2541) + (-572)$ $= -3113$		$\Sigma(\Delta H_c^\ominus \text{ products})$ $= -2878$

$$\begin{aligned}\Delta H^\ominus &= \Sigma(\Delta H_c^\ominus \text{ reactants}) - \Sigma(\Delta H_c^\ominus \text{ products}) \\ &= (-3113) - (-2878) \\ &= -235 \text{ kJ mol}^{-1}\end{aligned}$$

International Mindedness

An understanding of combustion enthalpies is useful in helping us to select fuels that are more climate conscious. While most of the world's energy supply comes from the combustion of hydrocarbons, scientists are actively optimising alternatives that deliver comparable energy in terms of efficiency, while emitting less harmful by-products. Scientific advances and political focus differ by country, though. How can we, as a global society, work together to mitigate the harmful effects of climate change from the combustion of hydrocarbons?

Hess's law and standard enthalpies of formation

In a nearly identical process, standard enthalpies of formation can be used as an application of Hess's law to determine the enthalpy of a reaction that cannot be measured directly.

$$\Delta H^\ominus = \Sigma(\Delta H_f^\ominus \text{ products}) - \Sigma(\Delta H_f^\ominus \text{ reactants})$$

This equation is the difference between the sum of product enthalpies of formation and the sum of the reactant enthalpies of formation. Like combustion, we must remember to multiply the standard enthalpies of formation by the stoichiometric balancing term (coefficient) in the balance equation before the enthalpies are summed.

Theory of Knowledge

Enthalpy of formation equations represent chemical reactions that do not necessarily take place in the natural world. Is it possible to use these equations as evidence to acquire further knowledge if they cannot be observed?

Consider the following example for determining the enthalpy for the combustion of propane using the standard enthalpies of formation (from [section 13](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/>) of the DP chemistry data booklet) of propane, carbon dioxide, and water. This example is also presented in a **Table 2** to assist with organising the data.

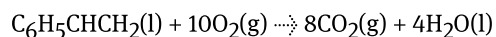
Table 2. Determining the enthalpy for the combustion of propane

	$\text{C}_3\text{H}_8(\text{g})$	+	$5\text{O}_2(\text{g})$	$\rightarrow$	$3\text{CO}_2(\text{g})$
	$\Delta H^\circ = ?$				
$\Delta H_f^\ominus$ (kJ mol ⁻¹)	-105		0		-394
Coefficient	1		5		3
Total enthalpy (kJ mol ⁻¹)	$= (-105) \times 1$ $= -105$		$= (0) \times 5$ $= 0$		$= (-394) \times 3$ $= -1182$
Sums (kJ mol ⁻¹)	$\Sigma(\Delta H_f^\ominus \text{ reactants})$ $= (-105) + (0)$ $= -105$				$\Sigma(\Delta H_f^\ominus \text{ products})$ $= (-1182)$ $= -1182$

$$\begin{aligned}\Delta H^\ominus &= \Sigma(\Delta H_f^\ominus \text{ products}) - \Sigma(\Delta H_f^\ominus \text{ reactants}) \\ &= (-1182) - (-105) \\ &= -1077 \text{ kJ mol}^{-1}\end{aligned}$$

Worked example 7

Using the standard enthalpies of formation from section 13 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/>) of the DP chemistry data booklet, calculate the enthalpy of reaction for the combustion of phenylethene:

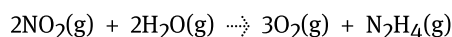


	$\text{C}_6\text{H}_5\text{CHCH}_2(\text{l})$	+	$10\text{O}_2(\text{g})$	-
$\Delta H_f^\ominus$ (kJ mol ⁻¹)	104		0	
Coefficient	1		10	
Total enthalpy (kJ mol ⁻¹)	104		0	
Sums (kJ mol ⁻¹)	$\Sigma(\Delta H_f^\ominus \text{ reactants})$ $= 104$			

$$\begin{aligned}
 \Delta H^\ominus &= \Sigma(\Delta H_f^\ominus \text{ products}) - \Sigma(\Delta H_f^\ominus \text{ reactants}) \\
 &= (-4296) - (104) \\
 &= -4400 \text{ kJ mol}^{-1}
 \end{aligned}$$

Worked example 8

Calculate the enthalpy for the reaction of nitrogen dioxide with water using the data provided.



$$\Delta H_f^\ominus \text{NO}_2 = 33 \text{ kJ mol}^{-1}$$

$$\Delta H_f^\ominus \text{H}_2\text{O}(\text{g}) = -242 \text{ kJ mol}^{-1}$$

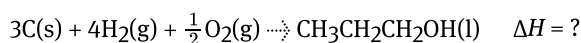
$$\Delta H_f^\ominus \text{N}_2\text{H}_4 = 48 \text{ kJ mol}^{-1}$$

	2NO ₂ (g)	+	2H ₂ O(g)	→
$\Delta H_f^\ominus$ (kJ mol ⁻¹)	33		-242	
Coefficient	2		2	
Total enthalpy (kJ mol ⁻¹)	66		-484	
Sums (kJ mol ⁻¹)	$\Sigma(\Delta H_f^\ominus \text{ reactants})$ $= 66 + (-484)$ $= -418$			

$$\begin{aligned}
 \Delta H^\ominus &= \Sigma(\Delta H_f^\ominus \text{ products}) - \Sigma(\Delta H_f^\ominus \text{ reactants}) \\
 &= (48) - (-418) \\
 &= 466 \text{ kJ mol}^{-1}
 \end{aligned}$$

Worked example 9

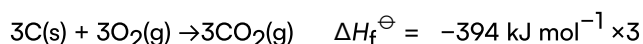
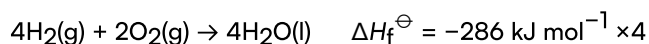
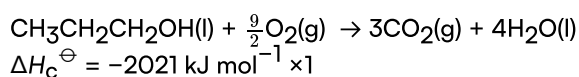
Calculate the enthalpy for the formation of propan-1-ol using the data provided.



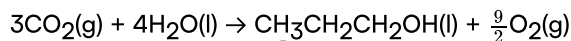
$$\Delta H_c^\ominus(\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}) = -2021 \text{ kJ mol}^{-1}$$

$$\Delta H_f^\ominus(\text{H}_2\text{O}(\text{l})) = -286 \text{ kJ mol}^{-1}$$

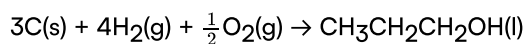
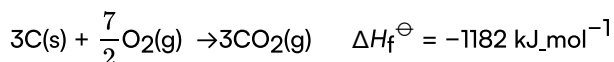
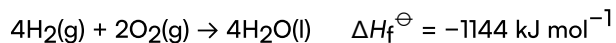
$$\Delta H_f^\ominus(\text{CO}_2) = -394 \text{ kJ mol}^{-1}$$



Therefore:



$$\Delta H_{\text{c}}^{\ominus} = +2021 \text{ kJ mol}^{-1}$$



$$\Delta H = 2021 - 1144 - 1182$$

$$= -305 \text{ kJ mol}^{-1}$$

Activity

- **IB learner profile attribute:** Communicator
- **Approaches to learning:** Research skills — Comparing, contrasting and validating information
- **Time required to complete activity:** 20 minutes
- **Activity type:** Group activity

In groups, work together to solve for the enthalpy of combustion of each of the fuels, listed below.

- Fuels
 - butane
 - propane
 - octane
 - ethanol
 - methane
- For each fuel type, one group member will be responsible for finding the enthalpy of formation for the reactants, another will be responsible for finding the enthalpy of formation of the products, and the group member will perform the calculation to determine the overall enthalpy.
- Make sure to rotate roles in the group for each question.